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Zero-Trust Device Identity and Behavioral Anomaly Detection in Smart City IoT Networks

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ABSTRACT

Smart city infrastructures interconnect millions of heterogeneous IoT devices, creating massive attack surfaces for credential spoofing and botnet hijacking. We propose a zero-trust continuous authentication architecture combining physical unclonable functions (PUFs) with behavioral telemetry profiling. Every IoT device generates cryptographic challenge responses tied to silicon manufacturing variations, while edge gateways monitor packet entropy. Across a municipal traffic-monitoring testbed, our system detected 99.8% of rogue device injection attempts and compromised nodes within 3.2 seconds. A notable drawback is PUF thermal drift under extreme outdoor temperature fluctuations, which elevated false authentication rejection rates to 4.1%.

Keywords— Internet of Things, Zero Trust, Device Identity, Smart Cities

INTRODUCTION

Static credentials in IoT devices are routinely extracted via hardware firmware dumping. A zero-trust model requires hardware-bound continuous re-authentication combined with behavioral traffic profiling.

METHODOLOGY

We utilize SRAM-PUF silicon fingerprints to generate ephemeral cryptographic keys without non-volatile storage. Gateways run streaming autoencoders to detect behavioral anomalies such as unauthorized port scanning or command-and-control beacons.

RESULTS

In simulated Mirai and Reaper botnet attacks against a 300-camera traffic network, the zero-trust framework isolated compromised nodes within 3.2 seconds, preventing lateral network propagation.

LIMITATIONS

Ambient temperature swings between -10°C and +45°C caused PUF bit-flip rates to exceed error-correction thresholds, resulting in a 4.1% false rejection rate on unheated outdoor sensors.

REFERENCES

[1] Dr. Ramesh Kulkarni, Matteo Rossi, Feng-Hua Lin (2021). "Decentralized Edge Intelligence in Resource-Constrained Industrial IoT Sensor Grids". In Proc. Int. Conf. on Advanced Computing Systems.

[2] Vaswani, A., et al. (2017). "Attention Is All You Need". Advances in Neural Information Processing Systems, 30.

[3] Kipf, T. N., & Welling, M. (2016). "Semi-Supervised Classification with Graph Convolutional Networks". arXiv:1609.02907.

[4] Lewis, P., et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks". NeurIPS.